ADS401 EVALUATION METHODS AND STATISTICS

From Usability Design to Statistical Evidence

Presentation 3: data collection, analysis, and interpretation

GROUP H

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The study evaluates one Museum AI system across four visitor tasks

TASK 01

Start

Open the guide and select an exhibit.

TASK 02

Ask

Ask one exhibit question and identify the answer.

TASK 03

Navigate

Move to the next recommended stop.

TASK 04

Recover

Handle an unclear or unsuitable answer.

12-15 XJTLU students or staff

10-question bilingual Wenjuanxing survey

Observation workbook and SPSS plan

Procedure: consent · warm-up · four tasks · Wenjuanxing and SUS · debrief
RQ1 task success · RQ2 usability and efficiency · RQ3 barriers and improvement

Two aligned sources share one anonymous Participant ID

Wenjuanxing survey

Role and prior experience
Task completion matrix
Task ease and confidence
SUS (10 items)
Content, navigation, recovery, trust, cognitive load
Open feedback

Observation sheet

Independent / assisted / failed
Time on task in seconds
Error count
Moderator help point
Anonymised think-aloud notes

Merge by Participant ID / verify rows / calculate scores / run tests / interpret with qualitative evidence

A short, pre-specified pipeline keeps the analysis reproducible

1

Export

Download Wenjuanxing Excel and observation sheet.

2

Merge

Match one survey row to one observation row by ID.

3

Code

Recode completion, reverse SUS items, and label themes.

4

Check

Missingness, duplicates, outliers, and skip logic.

5

Score

SUS, task success rate, time, errors, and theme counts.

SUS scoring

Odd items: score − 1 · Even items: 5 − score · Sum the ten adjusted items × 2.5

Task success

Independent = 1 · Assisted = 0 · Failed = 0 for the primary success rate

Each hypothesis has a matching test and effect interpretation

Question	Measure	Test	Interpretation
H1	Independent success, each task	Binomial test vs 80%	Proportion, exact p, 95% CI
H2	SUS	One-sample Wilcoxon vs 68; t-test if normal	Median/mean difference and effect size
H3	Ease, confidence, SUS, trust	Spearman correlations	Direction, strength, exploratory label
H4	Open feedback	Thematic coding	Frequency + anonymised examples

Check normality, report SUS alpha, exact p-values, effect sizes, and confidence intervals. Do not overstate causality from a small usability sample.

Final figures will be populated from the live questionnaire and observation sheet

Task success profile

Task 1 Start: __%

Task 2 Ask: __%

Task 3 Navigate: __%

Task 4 Recover: __%

Usability and perception

SUS: __ / 100

Average task ease: __ / 7

Average confidence: __ / 7

Trust: __ / 7

Cognitive load: __ / 7

SURVEY LIVE

No participant sessions are complete. No results are fabricated. Insert actual exported data before the final Presentation 3.

The final discussion connects evidence to design decisions

01 Task success identifies where the Museum AI system fails or needs help.

02 SUS and task ratings show whether the interaction feels learnable and controllable.

03 Open responses explain the barriers behind the numbers.

04 Recommendations target onboarding, content clarity, navigation, and recovery.

Q&A

Limitations: small convenience sample, controlled setting, one system version, and selected tasks.

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